NISHA KOLEKAR

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SUMMARY

Data and AI professional with a proven track record in Python, SQL, machine learning, and statistical modeling. Successfully reduced manual effort by 90% using AI-based automation for question paper generation, and improved manufacturing yield by identifying the drivers of casting rejection through data analysis. Currently pursuing an MS in Artificial Intelligence Engineering at Arizona State University, committed to turning complex data into clear insights that drive sound business decisions.

TECHNICAL SKILLS

Languages & Querying: Python, SQL (PostgreSQL, MySQL, Oracle), R, Java, Go

Analytics & Modeling: machine learning, pandas, NumPy, scikit-learn, statistical modeling, regression, Random Forest, gradient boosting (XGBoost, CatBoost), data mining, data cleaning and preprocessing, model evaluation (R^2, MAE, accuracy)

Visualization & Reporting: data visualization (matplotlib, seaborn), exploratory data analysis, quantitative analysis

AI / ML: Large language models, RAG, agentic AI, LangChain, prompt engineering, Hugging Face, Ollama, NLP, deep learning

AI Tooling: Claude, AI-assisted development and analysis workflows

Platforms & Engineering: Microsoft Azure, AWS (Elastic Beanstalk), Flask, REST APIs, Spring Boot, Git, Jenkins

EDUCATION

Arizona State University, Tempe, AZ — *MS, Artificial Intelligence Engineering (Software Engineering)* | Aug 2026 – Present

Sanjay Ghodawat University, Kolhapur, India — *B.Tech, Computer Science and Engineering*, GPA 8.62/10 | Aug 2017 – Aug 2021

EXPERIENCE

Senior Associate – Learning, *Infosys Limited*, Mysuru, India — Apr 2024 – Jan 2025

- Scaled an LLM-based question paper generation system to support multiple languages and new features, expanding its user base across training cohorts.
- Built a chatbot on Llama 3.1-8B with LangChain and prompt engineering, and evaluated response quality across prompting strategies to select the configuration that performed best.
- Collaborated cross-functionally with content, training and engineering stakeholders to translate business requirements into working systems and measurable improvements.
- Delivered technical training on AI, Java, Spring and Go, communicating complex technical concepts to non-technical audiences; monitored feedback metrics across cohorts and sustained a 4.8/5 average rating.

Systems Engineer, *Infosys Limited*, Mysuru, India — Mar 2022 – Apr 2024

- Co-developed an AI question paper generation system on Microsoft Azure (GPT-3.5), **reducing generation time by 90%** and removing a recurring manual bottleneck for the training organization.
- Designed and tested prompting strategies — system, few-shot and role prompting — monitoring output quality metrics across iterations to determine which approach produced reliable results at scale.
- Queried and validated relational data (Oracle, PostgreSQL) backing the platform, and built REST services in Java and Spring Boot to integrate model output into existing workflows.

Data Science Intern, *Menon and Menon Ltd.*, Kolhapur, India — Feb 2021 – May 2021

- Analyzed manufacturing process data to identify the drivers of casting rejection, scoping the problem directly with plant stakeholders before modeling.
- Cleaned and prepared the dataset — deduplication, missing-value handling, normalization — and used exploratory analysis and matplotlib visualizations to surface patterns for the business team.
- Built and compared multiple machine learning models; Random Forest performed best at approximately 87% accuracy.
- Translated model findings into an optimized casting recipe that **reduced the rejection rate and improved yield**, turning the analysis into an operational change.

PROJECTS

AI Data Center Sustainability & Design Assistant — *Graduate team project, ASU* — 2026 – Present

- Building an agentic AI system that models U.S. data center energy, water, carbon and noise impact, scoring cooling and siting alternatives on PUE and WUE while documenting assumptions and uncertainty. Currently sourcing facility, climate and utility datasets.

Student Performance Prediction — End-to-End ML — *github.com/Nisha2306/mlproject* — Personal

- Built a full pipeline from exploratory analysis to deployment: preprocessing with scikit-learn, comparison of seven regression models (best R^2 0.88, MAE 4.2), and a Flask prediction service configured for AWS Elastic Beanstalk.

CERTIFICATIONS & AWARDS

Infosys Certified Generative AI Professional – Expert • Infosys Certified Applied Generative AI Professional • Microsoft Technology Associate: Programming using Python • NPTEL: Python for Data Science • Infosys Certified Spring REST Professional • **Infosys Insta Award** (Dec 2023), Go content and AI initiatives.